

EXCEPTION HANDLING

Errors:

A programming mistake is said to be an error.

There are three types of errors:

- 1) Compile time errors (Syntax errors)
- 2) Run time errors (Exceptions)
- 3) Logical errors

Compile time errors are also called syntax errors.

Run time errors are also called exceptions.

Example of syntax error:

```
i=0
if i==0
    print(i)
```

In the above example we have missed the : symbol after if statement. It is called as syntax error.

Example of exception:

```
a=10
b=0
print(a//b)
```

In the above example ZeroDivisionError occurs. It is called as an exception.

This type of exceptions can be handled explicitly to display user friendly error messages.

Handling Exceptions: In exception handling, we use the following keywords:

1) try 2) except 3) raise 4) finally

try block is used to write the task code.

except block is used to write user friendly error message.

raise keyword used to raise the exception.

finally block is used to perform cleanup activities.

Clean up activities are closing a file, closing a database connection, closing a network socket, ..etc.,

Whenever exception occurs in a program python interpreter raise the exception, passed to except block and except block code is executed.

Syntaxes:

1) try with except block:

try:

=====

except ExceptionClassName:

=====

2) try with multiple except blocks:

try:

=====

except ExceptionClassName1:

=====

except ExceptionClassName2:

=====

3) Handling multiple exceptions with single except block:

try:

```
=====
except(ExceptionClassName1, ExceptionClassName2, .....)
```

4) Default except block to handle all exceptions:

```
try:
=====
except:
=====
```

Examples:

1)

try:

```
a=int(input("Enter First Number: "))
b=int(input("Enter Second Number: "))
c=a//b
print(c)
```

except ZeroDivisionError:

```
print("Enter Second Number Except Zero")
```

2)

try:

```
a=int(input("Enter First Number: "))
b=int(input("Enter Second Number: "))
c=a//b
print(c)
```

except ValueError:

```
    print("Enter Numbers Only")
except ZeroDivisionError:
    print("Enter Second Number Except Zero")
3)
try:
    a=int(input("Enter First Number: "))
    b=int(input("Enter Second Number: "))
    c=a//b
    print(c)
except(ZeroDivisionError, ValueError):
    print("Enter two numbers and second number except zero")
4)
try:
    a=int(input("Enter First Number: "))
    b=int(input("Enter Second Number: "))
    c=a//b
    print(c)
except:
    print("Enter two numbers and second number except zero")
```

Program to create user defined exception:

```
class NegativeNumberException(Exception):
    pass

class Test:
    def cube(self, a):
```

```
    if a>0:
        print(a*a*a)
    else:
        raise NegativeNumberException

try:
    t=Test()
    x=int(input("Enter any number: "))
    t.cube(x)
except NegativeNumberException:
    print("Enter positive number only")
```

Program to demonstrate finally block:

```
f=None
try:
    file=input("Enter any file name: ")
    f=open(file, "r")
    x=f.read()
    print(x)
except FileNotFoundError:
    print("Enter existed file")
finally:
    if f!=None:
        f.close()
```

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